

FINITENESS OF PLANAR FIVE-BODY CENTRAL CONFIGURATIONS FOR GENERIC EQUAL-PAIR MASSES

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ABSTRACT. For a nonempty relative Zariski-open set of positive triples (a, b, c) , the planar five-body problem with masses (a, a, b, b, c) has finitely many central configurations up to similarity. This family lies in the exceptional mass set of Albouy and Kaloshin. We prove the result by excluding cells of a tropical prevariety with exact Laurent identities, elimination, and modular computations that check entire cone stars. The remaining cells lie in an open half-space. Fixed-potential augmentation and the finiteness of the set of potential values then give the physical finiteness statement.

1. INTRODUCTION

A configuration of n point masses $m_i > 0$ at positions $x_i \in \mathbb{R}^2$ is a *central configuration* (CC) if

$$\sum_{j \neq i} \frac{m_j(x_j - x_i)}{|x_j - x_i|^3} = -\lambda(x_i - x_{\text{cm}}) \quad \text{for all } i,$$

where x_{cm} is the center of mass and $\lambda > 0$. Smale's sixth problem asks whether, for every choice of positive masses, there are only finitely many CCs up to similarity. Hampton and Moeckel settled the case $n = 4$ [2]. Albouy and Kaloshin [1] settled $n = 5$ for all masses outside an explicit codimension-two subvariety $\mathcal{S}$ of the mass space. In Section 8.6, equation (32) of [1] is precisely $m_1 = m_2$ and $m_3 = m_4$. Their Theorem 6, proof part (ii), adds this locus and its relabelings to the exceptional set. Parts (i), (iii), and (iv) add other codimension-two mass loci; the equal-pair locus is not the entire positive-mass exceptional set. Our theorem concerns a relative-generic subset of this one family.

Jensen and Leykin [3] gave a tropical proof for generic masses in the ambient mass space. Their equal-pair valuation experiments in Section 4.2 do not yield pointed recession cones. Equal valuations allow unequal masses with the same leading order; here we impose the exact identities $m_1 = m_2$ and $m_3 = m_4$ in the coefficient field. The additional algebraic exclusions, rather than the support fan alone, remove the obstruction to pointedness.

Moczurad and Zgliczyński [5] certify counts at selected exceptional mass choices, including choices in the same equal-pair family, and discuss persistence in small neighborhoods. Their investigation of this family concerns specific mass choices rather than a relative Zariski-open subset of the family.

The all-equal planar case was already classified by Moczurad and Zgliczyński [7]; Lee and Santoprete [8] had earlier classified isolated equal-mass solutions. These results concern a special point of the mass family. Results on kite configurations [10], symmetric stacked configurations [11], and trapezoidal configurations with two equal mass pairs [12] impose geometric restrictions. The mass equalities in the present theorem do not impose reflection symmetry or a prescribed shape on the central configurations.

Hampton and Jensen [9] previously used polyhedral initial-ideal computations and elimination to prove spatial five-body finiteness with explicit mass exceptions. Chang and Chen [13, 14] develop symbolic diagram and mass-relation methods toward planar six-body finiteness; their combined preprint [15], Section 10.6, retains Diagram 19 for the mass relation $m_1 = m_2$, $m_4 = m_5$, equivalent to our family after relabeling. Those equalities already hold here, so they cannot exclude

that diagram. Our method instead proves emptiness of distance-initial systems within the equal-pair family, using fixed-potential identities as additional equations; it does not identify individual support cones with their diagrams. Recent exact replication and tropical certificate screening by Santib  n  ez-Leal [16] concern ambient-generic results and valuation experiments. These works are precedents for the algebraic and verification techniques used here. Wang and Zhao [6, Theorem 8.1 and Corollary 8.2] obtain perturbative finiteness for non-scalar balanced configurations under a central-configuration finiteness hypothesis; their five-body corollary uses masses outside the Albouy–Kaloshin exceptional set.

Our contribution is a relative-generic finiteness result within the equal-pair exceptional mass family, allowing all real planar configurations without a geometric symmetry assumption. The argument extends the known ambient-generic results into this family. The result gives neither all-mass finiteness nor exact counts.

Theorem 1. *There is a nonempty Zariski-open subset $\mathcal{O} \subset \{(a, b, c) \in \mathbb{R}_{>0}^3\}$ such that for every $(a, b, c) \in \mathcal{O}$, the planar five-body problem with masses (a, a, b, b, c) has finitely many central configurations up to similarity. Here similarity identifies translations, rotations, reflections, and positive changes of scale; body labels are retained.*

2. EQUATIONS AND TROPICAL SET-UP

We use coordinates r_{ij} , $1 \leq i < j \leq 5$, in lexicographic pair order, with $r_{ji} = r_{ij}$ and $r_{ii} = 0$. Normalize the multiplier to $\lambda = M = \sum_i m_i$ and set $S_{ik} = r_{ik}^{-3} - 1$ for $i \neq k$. The ordered distance equations are

$$(1) \quad A_{ij} = \sum_{k \neq i} m_k S_{ik} (r_{jk}^2 - r_{ik}^2 - r_{ij}^2) = 0, \quad i \neq j.$$

There are $5 \cdot 4 = 20$ equations, because the first index specifies the force vector being tested. We also use the ten redundant symmetric equations $A_{ij} + A_{ji} = 0$, $i < j$, and the five determinants

$$(2) \quad \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & r_{ij}^2 & r_{ik}^2 & r_{il}^2 \\ 1 & r_{ij}^2 & 0 & r_{jk}^2 & r_{jl}^2 \\ 1 & r_{ik}^2 & r_{jk}^2 & 0 & r_{kl}^2 \\ 1 & r_{il}^2 & r_{jl}^2 & r_{kl}^2 & 0 \end{pmatrix} = 0, \quad i < j < k < \ell.$$

Multiply A_{ij} by $\prod_{k \neq i} r_{ik}^3$, and each symmetric equation by the product of the corresponding two denominators. Removing common distance monomials and nonzero rational scalars gives the stored system $F = (f_1, \dots, f_{35})$: the twenty ordered equations, the ten symmetric equations, then the determinants. Within each block the indices are lexicographic. These removals are units on the distance torus; no mass factor is removed. The formula comparison in `companion/checkers/verify_equations.py` reconstructs all thirty-five polynomials from (1)–(2) and matches `data/sys.json` exactly up to those units. The formulas agree with [3, Section 2.2]. The symmetric sums are redundant algebraically, but their support hypersurfaces can strengthen the tropical prevariety.

Lemma 2 (Physical meaning of the distance equations). *For positive masses and a collision-free real planar configuration, its positive mutual distances satisfy the twenty equations (1) if and only if the configuration is central with multiplier M . This includes collinear configurations. Every such distance tuple satisfies (2) and hence F .*

Proof. Put $q_i = \sum_{k \neq i} m_k S_{ik} (x_k - x_i)$. The cosine identity gives $A_{ij} = -2q_i \cdot (x_j - x_i)$. Since q_i belongs to the real span of the vectors $x_j - x_i$, vanishing of all A_{ij} forces $q_i = 0$. This argument

holds when that span has dimension one. Moreover,

$$q_i = \sum_{k \neq i} \frac{m_k(x_k - x_i)}{r_{ik}^3} + M(x_i - x_{\text{cm}}).$$

Thus $q_i = 0$ is exactly the normalized central-configuration equation, and the converse follows from the same identity. The determinants vanish for four points in the plane. $\square$

These are scalar-product equations, not area-factor equations that vanish identically on a line. We require the equivalence only on real Euclidean distance tuples; arbitrary complex zeros of F need not represent physical configurations.

Each ordered or symmetric polynomial is homogeneous of degree one in the masses; the determinants do not involve masses. We may therefore divide all masses by $c > 0$ and take $c = 1$. Write $k_0 = \mathbb{Q}(a, b)$, $K = \bar{k}_0$, with trivial valuation, and $X = V(F) \subset (K^*)^{10}$.

Fixed potential.

The normalization above is $\lambda = M$, where $M = \sum_i m_i$. If $I = \sum_i m_i |x_i - x_{\text{cm}}|^2$, then $MI = \sum_{i < j} m_i m_j r_{ij}^2$, and the virial identity is $U = MI$, with $U = \sum_{i < j} m_i m_j / r_{ij}$. Define

$$G = \sum_{i < j} m_i m_j r_{ij}^2 - U,$$

$$H = \sum_{i < j} m_i m_j \prod_{(k,l) \neq (i,j)} r_{kl} - U \prod_{k < l} r_{kl}.$$

Here U is an additional torus coordinate. For a fixed $u \neq 0$, write G_u, H_u for $U = u$ and put $X'_u = V(F, G_u, H_u)$. Every real normalized central configuration of potential u lies in X'_u . Lemma 8 reduces the physical problem to finitely many such nonzero potential values. Our final pointedness argument applies to these augmented varieties, and does not assert finiteness of the full complex variety $V(F)$.

Prevariety fan.

Let Σ be the polyhedral fan computed by `gfan_tropicalintersection` from the supports of $f_1, \dots, f_{35}$ (max convention; file `data/fan.out`). It has f-vector $(1, 1603, 6110, 6525, 2200)$, 16,438 nonzero cones and trivial lineality space. We have $\text{Trop}(X) \subseteq \bigcap_i \text{Trop}(V(f_i)) = |\Sigma|$.

3. GEOMETRIC AND SPECIALIZATION LEMMAS

Throughout, $\text{in}_w f$ is the max-initial form: the sum of the terms of f whose exponents maximize $\langle w, \cdot \rangle$. $\text{Trop}(X)$ is the set of $w \in \mathbb{R}^{10}$ such that $\text{in}_w I(X)$ contains no monomial. By the Fundamental Theorem [4, Thm. 3.2.3] and the Structure Theorem [4, Thm. 3.3.5], this is the tropicalization over the trivially valued algebraically closed field K . We use the max convention throughout; the min convention in [4] transfers by $w \mapsto -w$. For a cone $C = \text{cone}(v_1, \dots, v_s) \in \Sigma$ we put $w_C = \sum_j v_j$.

Lemma 3 (Cell constancy). *Let f be a polynomial whose initial form is used, with cone rays extended by a zero U -coordinate when necessary, and let*

$$\mathcal{F}_f = \bigcap_j \arg \max_{m \in \text{supp } f} \langle v_j, m \rangle.$$

If $\mathcal{F}_f \neq \emptyset$, then $\text{in}_w f = \text{in}_{w_C} f$ for every w in the relative interior of C .

Proof. Write $w = \sum \lambda_j v_j$ with all $\lambda_j > 0$. For $m_0 \in \mathcal{F}_f$ and $m \in \text{supp } f$, each $\langle v_j, m_0 - m \rangle \geq 0$. Hence $\langle w, m_0 - m \rangle \geq 0$, with equality iff every term vanishes, i.e. iff $m \in \mathcal{F}_f$. $\square$

The verifier checks the hypothesis of Lemma 3 for every generator used in every certificate.

Lemma 4 (Exclusion by a unit certificate). *Let Φ be a polynomial system in n torus coordinates r , over $\mathbb{Q}[a, b]$, and let $C \subset \mathbb{R}^n$ be a rational polyhedral cone. Choose $\mathcal{P} \subseteq \Phi$ satisfying Lemma 3 on C . Let $\mathcal{V}$ be the common homogeneity space of the forms $\text{in}_{w_C} f$, $f \in \mathcal{P}$, and let J be coordinates such that $\mathcal{V} \rightarrow \mathbb{Q}^{\mathcal{D}}$ is surjective. Denote by $\bar{f}$ the form obtained by fixing $r_j = 1$ for $j \in \mathcal{D}$. Suppose*

$$\sum_{f \in \mathcal{P}} C_f \bar{f} + C_0 \left(1 - \zeta \prod_{j \notin \mathcal{D}} r_j\right) = \ell, \quad 0 \neq \ell \in \mathbb{Q}[a, b],$$

where all cofactors are in $\mathbb{Q}[a, b][\langle r_j \rangle_{j \notin \mathcal{D}}, \zeta]$. Then $\text{relint}(C) \cap \text{Trop}(V(\Phi)) = \emptyset$ over K . The identity remains an exclusion after specializing masses whenever $\ell \neq 0$ and the used initial supports are preserved.

Proof. If the initial ideal at $w \in \text{relint}(C)$ contains no monomial, its subideal generated by the chosen initial forms is proper in $K[r^{\pm 1}]$. The Nullstellensatz gives a common torus zero z . Integer vectors in $\mathcal{V}$ act on these zeros by coordinatewise torus scaling. The surjective projection onto $\mathcal{D}$ supplies an integer matrix with nonzero determinant on those coordinates. Over the algebraically closed field K , the corresponding isogeny of tori is surjective, so a zero can be moved to the slice $r_j = 1$, $j \in \mathcal{D}$. Substituting this zero and $\zeta = (\prod_{j \notin \mathcal{D}} r_j)^{-1}$ into the identity gives $0 = \ell$, a contradiction. The same identity proves the specialized assertion. $\square$

We use the lemma both in ten distance coordinates and in eleven coordinates (r, U) , extending each distance weight by a zero U -coordinate. The auxiliary variable ζ is distinct from the fixed potential value u .

Lemma 5 (Exclusion from one modular specialization). *Let $\Phi = F \cup \{G, H\}$ as defined in Section 2, with U the extra coordinate of weight 0. For a cone $D \in \Sigma$ let Φ_D be the set of $f \in \Phi$ that satisfy Lemma 3 on D and whose form $\text{in}_{w_D} f$ does not involve U . Let $C \in \Sigma$ with $F \subseteq \Phi_C$. Suppose that there are a prime p and $(a_0, b_0) \in \mathbb{Z}^2$ such that*

- (1) *every coefficient (an element of $\mathbb{Z}[a, b]$) of every form $\text{in}_{w_C} f$, $f \in \Phi_C$, is nonzero modulo p at (a_0, b_0) , and*
- (2) *for every cone $D \in \Sigma$ having C as a face (including $D = C$), the forms $\text{in}_{w_D} f$, $f \in \Phi_C \cap \Phi_D$, specialized at (a_0, b_0) and reduced modulo p , have no common zero in the torus over $\mathbb{F}_p$.*

Then the forms $\text{in}_{w_C} f$, $f \in \Phi_C$, have no common zero in the torus over $K = \overline{\mathbb{Q}(a, b)}$. Consequently, the cone is excluded for every augmented fixed-potential variety X'_u , $u \neq 0$, at generic masses.

Proof. Write $h \in \mathbb{Z}[a, b]$ as $h = \sum \gamma_{ij} (a - a_0)^i (b - b_0)^j$ with $\gamma_{ij} \in \mathbb{Z}$, and put $\nu(h) = \min\{v_p(\gamma_{ij}) + i\sqrt{2} + j\sqrt{3}\}$. Since $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$, the minimum is attained by exactly one term, and this term is multiplicative. So ν extends to a valuation of $\mathbb{Q}(a, b)$. Its residue field is $\mathbb{F}_p$, and $\nu(h) = 0$ iff $h(a_0, b_0) \not\equiv 0 \pmod{p}$. Extend ν to K (residue field $\mathbb{F}_p$, value group $\Gamma \subseteq \mathbb{R}$ divisible), and fix a splitting $\lambda \mapsto t^\lambda$ of ν on Γ [4, Lemma 2.1.15].

Suppose that z is a common zero in the torus over K , and write $z = t^{-w}y$ with $w \in \Gamma^{11}$, $\nu(y_i) = 0$. Because the forms do not involve U , we may replace the U -entry of w by 0. Let $g = \text{in}_{w_C} f = \sum c_m r^m$ with $f \in \Phi_C$. By (1), every $\nu(c_m) = 0$. Multiplying $g(z) = 0$ by $t^{h_{\max}}$, with $h_{\max} = \max_m \langle w, m \rangle$, and reducing gives $\overline{\text{in}_w g}(\bar{y}) = 0$. Here the bar denotes specialization at (a_0, b_0) modulo p . For $\varepsilon > 0$ small enough (uniformly for the finitely many f), $\text{in}_w \text{in}_{w_C} f = \text{in}_{w'} f$ with $w' = w_C + \varepsilon w$. If $w' \notin |\Sigma|$, then some $f \in F \subseteq \Phi_C$ has a monomial $\text{in}_{w'} f$. Its specialized coefficient is nonzero, so it cannot vanish at $\bar{y}$, a contradiction. Otherwise $w' \in \text{relint } D$ for some $D \in \Sigma$. Since ε is small and $w_C \in \text{relint } C$, C is a face of D . For $f \in \Phi_C \cap \Phi_D$, Lemma 3 gives $\text{in}_{w'} f = \text{in}_{w_D} f$. Thus $\bar{y}$ is a common torus zero of the specialized forms $\text{in}_{w_D} f$, $f \in \Phi_C \cap \Phi_D$, contradicting (2). The final statement follows from the unit ideal over k_0 and Lemma 12 below; its generators do not involve U , so the mass condition is uniform in u . $\square$

The valuation and initial-degeneration principles underlying Lemma 5 are standard; see [17, Theorem 5.6] and [4]. The lemma packages them into a sufficient criterion for this computation. Lemma 5 turns a modular computation into a proof. It needs *no* computation over $\mathbb{Q}(a, b)$, only that no escape to the boundary of the cell is possible modulo p , which (2) states as exclusion of every cone of the star of C . Condition (2) is checked by modular cofactor identities and exact elimination replays, orchestrated by `verifier/verify_lemS.py`. The target-cell replays and their exceptional branches are described in Section 5.

Remark 6 (Why the star condition is needed). Emptiness of a specialized fiber can coexist with generic torus solutions even when every coefficient survives. For example, $f_1 = x + y$ and $f_2 = x + (1 + a)y + 1$ have no common zero at $a = 0$, but have the torus solution $(x, y) = (1/a, -1/a)$ for $a \neq 0$. For the neighboring weight $(1, 1)$, the specialized max-initial forms both become $x + y$, which has torus zeros. The additional star check detects this escape to infinity.

Lemma 7 (Residual half-space implies finiteness). *Let $W \subset (K^*)^n$ be an algebraic variety over a trivially valued algebraically closed field of characteristic zero. Let Σ be a finite polyhedral fan with $\text{Trop}(W) \subseteq |\Sigma|$. Let $\mathcal{E}$ be a collection of nonzero cones such that $\text{relint}(E) \cap \text{Trop}(W) = \emptyset$ for every $E \in \mathcal{E}$. Suppose each cone of $\Sigma \setminus \mathcal{E}$ is generated by its listed rays, and let $\mathcal{R}$ be the rays of these remaining cones. If $\eta \in \mathbb{Z}^n$ satisfies $\langle \eta, v \rangle > 0$ for every $v \in \mathcal{R}$, then W is finite.*

Proof. Every nonzero point of a remaining cone is a nonnegative combination of its rays, so $\langle \eta, w \rangle > 0$ for all nonzero $w \in \text{Trop}(W)$. For an irreducible component Y and the monomial map $r \mapsto r^\eta$, the tropicalization of the image closure is the linear image $\langle \eta, \text{Trop}(Y) \rangle$ [4, Corollary 3.2.13]. A closed irreducible subset of the one-dimensional torus is a point or the whole torus. Its tropicalization is respectively $\{0\}$ or $\mathbb{R}$, and the latter cannot be contained in $[0, \infty)$. Thus $\langle \eta, w \rangle = 0$ on $\text{Trop}(Y)$, forcing $\text{Trop}(Y) = \{0\}$. The dimension theorem [4, Theorem 3.3.5] gives $\dim Y = 0$. There are finitely many components. $\square$

Lemma 8 (Finitely many physical potential values). *Fix positive real masses. Among collision-free real planar central configurations normalized by $\lambda = M$, the potential takes only finitely many values, all positive. Every such configuration satisfies $U = MI = \sum_{i < j} m_i m_j r_{ij}^2$.*

Proof. Work over $\mathbb{C}$ on the center-of-mass-zero position space, restricted to configurations with all squared mutual distances nonzero. Adjoin nonzero r_{ij} with $r_{ij}^2 = (x_i - x_j) \cdot (x_i - x_j)$. This distance cover is smooth and étale over that open position space, since each derivative $2r_{ij}$ is invertible. The physical configurations use its positive real distance branches.

On this smooth cover put $f = U + \frac{M}{2}I$. Differentiating U in the position directions shows that the normalized force equations are exactly $df = 0$. The center-of-mass constraint loses no equation: the weighted sum of the vectors $\sum_{j \neq i} m_j(x_j - x_i)/r_{ij}^3 + Mx_i$ is zero, so a critical point on the constrained space satisfies the full force equations. Thus the normalized central configurations belong to the algebraic critical locus of f .

On every irreducible component of that locus, the differential of the restricted function f is zero on its dense smooth locus. In characteristic zero this makes f constant. Dotting the force equations with $m_i x_i$ and summing gives $U = MI$, so there $f = \frac{3}{2}U$. The critical locus has finitely many irreducible components, hence finitely many potential values on its physical points. Finally $MI = \sum_{i < j} m_i m_j r_{ij}^2$, and positivity gives $U > 0$. This is the physical version of [1, Lemma 2]. $\square$

Lemma 9 (The scaling ray and other negative directions). *Fix positive real masses, let $u \in \mathbb{C}^*$ and let $G_u = \sum_{i < k} m_i m_k r_{ik}^2 - u$. Put $X_u = V(F \cup \{G_u\}) \subseteq (\mathbb{C}^*)^{10}$. Every real planar central configuration lies in some X_{u_j} with $u_j \neq 0$. If $w \in \mathbb{R}^{10}$ has all coordinates negative, then $\text{in}_w G_u = -u$ is a unit, so $w \notin \text{Trop}(X_u)$. In particular the scaling ray $\mathbb{R}_{>0}(-1, \dots, -1)$ (cell 79) is excluded for every X_u . More generally, a cone on whose relative interior $\text{in}_w G_u$ is a single fixed monomial is excluded (96 cells before overlap with other exclusions).*

Proof. Real configurations have $U > 0$, so $u_j = U \neq 0$. The monomials of G_u have w -weights $2w_{ik} < 0$ and the constant has weight 0. $\square$

Remark 10 (Certificates for the augmented system). Some certificates are for $Y = V(F \cup \{G\}) \subseteq (\mathbb{C}^*)^{11}$, where $G = \sum_{i < k} m_i m_k r_{ik}^2 - U$ and U is an extra torus coordinate. For $u \neq 0$ the map $r \mapsto (r, u)$ embeds X_u into Y . With the trivial valuation, the U -coordinate has valuation 0, so $\text{Trop}(X_u) \times \{0\} \subseteq \text{Trop}(Y)$. Hence if a unit certificate (Lemma 4, with Lemma 3 applied in $\mathbb{R}^{11}$) shows that $(w, 0) \notin \text{Trop}(Y)$ for all $w \in \text{relint } C$, then $\text{relint } C \cap \text{Trop}(X_u) = \emptyset$ for every $u \neq 0$. The cone's rays are extended by a 0 in the U -coordinate, and constancy of the initial forms is checked on the cone $C \times \{0\}$. Such certificates are just as good as certificates for X in the proof below, since the proof only uses the X_{u_j} .

Remark 11 (Certificates using the definition of U). Let

$$\begin{aligned} H &= \sum_{i < k} m_i m_k \prod_{(j,l) \neq (i,k)} r_{jl} - U \prod_{j < l} r_{jl} \\ &= \left(\sum_{i < k} \frac{m_i m_k}{r_{ik}} - U \right) \prod_{j < l} r_{jl}, \end{aligned}$$

which is the denominator-cleared potential identity (file `verifier/verify.py`, `augmentH`; index 36). For $u \neq 0$ put $X'_u = V(F \cup \{G_u, H_u\}) \subseteq X_u$. A real planar central configuration with potential $U = u$ satisfies $H_u = 0$, because $U = \sum m_i m_k / r_{ik}$ is its potential. So it lies in X'_u , and the real central configurations lie in $\bigcup_j X'_{u_j}$. As in Remark 10, $\text{Trop}(X'_u) \times \{0\} \subseteq \text{Trop}(Y')$ with $Y' = V(F \cup \{G, H\}) \subseteq (\mathbb{C}^*)^{11}$, and $\text{Trop}(X'_u) \subseteq \text{Trop}(X_u)$. Hence certificates for F , for $F \cup \{G\}$ and for $F \cup \{G, H\}$ (header `system F+G+H`, directory `exact/hcert/`) all exclude their cones from $\text{Trop}(X'_u)$, and the proof of the theorem goes through with X'_{u_j} in place of X_{u_j} .

Lemma 12 (Spreading exclusions to mass parameters). *Consider finitely many initial systems over $\mathbb{Q}[a, b]$, each with no torus zero over $\overline{\mathbb{Q}(a, b)}$. There is a nonzero polynomial $q \in \mathbb{Q}[a, b]$ such that all these systems have no torus zero after any characteristic-zero specialization with $q(a, b) \neq 0$. A single such q can also preserve every support of a specified finite polynomial system.*

Proof. For each initial system, add $1 - \zeta \prod r_j$ and apply the Nullstellensatz. The resulting ideal is the unit ideal over $\overline{\mathbb{Q}(a, b)}$, hence over $\mathbb{Q}(a, b)$ by faithful scalar extension. Clear the finitely many mass denominators in a unit identity to obtain a nonzero polynomial $q_i \in \mathbb{Q}[a, b]$. Multiplying these q_i and all nonzero coefficient polynomials of the specified system gives q . For $q \neq 0$, the identities specialize and no coefficient vanishes, so the supports are unchanged. Equivalently, the parameter set where a torus zero exists is constructible and misses the generic point; its closure is proper. The identities give a direct proof of that properness. $\square$

For augmented systems, the explicit certificates have coefficients in $\mathbb{Q}(a, b)$ and treat U as a torus variable. The structural and modular exclusions use U -free forms. Apply the lemma to these systems (including U as a coordinate when needed). The resulting mass polynomial is independent of u , so one open set works for every fixed $u \neq 0$. The coefficients of G_u, H_u also preserve their distance supports for positive masses and $u \neq 0$.

Proof of Theorem 1 from the certificates. First work at generic masses over $\overline{\mathbb{Q}(a, b)}$. The identities, elimination replays, and Lemma 5 exclude the certified cones for every X'_u , $u \neq 0$. In particular, the forms used by Lemma 5 do not involve U , so this exclusion is independent of the chosen nonzero potential value.

Lemma 12, the denominators of the explicit identities, and the coefficient polynomials of F give a hypersurface $B = V(q) \subset \mathbb{C}^2$, with $0 \neq q \in \mathbb{Q}[a, b]$. Outside B , the support fan and all exclusions remain valid, uniformly in $u \neq 0$.

The coverage and half-space checks in Section 4 therefore make every X'_u finite by Lemma 7. Lemma 8 supplies finitely many positive potential values for the real central configurations. Their distance tuples lie in a finite union of these varieties. A labeled real distance tuple determines a configuration up to Euclidean isometry, including reflection, and $\lambda = M$ fixes scale. Finally, a nonzero polynomial cannot vanish on the entire positive quadrant, so $\mathbb{R}_{>0}^2 \setminus B$ is nonempty. Pull it back under $(a, b, c) \mapsto (a/c, b/c)$; after clearing powers of c , this gives $\mathcal{O} = \{(a, b, c) \in \mathbb{R}_{>0}^3 : Q_*(a, b, c) \neq 0\}$ for a nonzero homogeneous $Q_* \in \mathbb{Q}[a, b, c]$. Thus $\mathcal{O}$ is the positive-mass complement of a hypersurface defined over $\mathbb{Q}$. □

Remark 13. Explicit Laurent certificates record mass denominators. For exclusions established only by structural elimination or Lemma 5, the additional mass polynomials in Lemma 12 are proved to exist but are not all extracted. Thus the modular point $(3, 7)$ verifies the specified stars, not every exclusion at that named mass pair. An exact scan of 1,662 recorded cofactor and split-tree files finds 506 distinct denominator expressions that vanish at $(3, 7)$, spread over ten files; see `data/mass_denominator_audit.json`. Those recorded identities cannot be specialized there as written. This does not establish failure of finiteness at that point; obtaining a named instance requires replacement identities and replay of the remaining exclusions.

4. COMPUTATION

Symmetry.

The swaps $(1\ 2)$ and $(3\ 4)$ preserve the polynomial system and the fan, giving 4,781 orbits under this order-four group of the 16,438 nonzero cones. The relabeling $\sigma = (1\ 3)(2\ 4)$, together with $a \leftrightarrow b$, supplies an additional symmetry. The generated group has order eight. The scripts `symcheck.py` and `symcheck2.py` reconstruct and check the polynomial and fan actions; transported identities are checked again against their target cones.

Search and exact certificates.

Initial systems were first screened at mass values modulo 32003. Such a unit test was used to choose subsets and computation orders. The direct characteristic-zero pipeline, using SINGULAR [19], then computed unit ideals over $\mathbb{Q}(a, b)$, extracted lift cofactors, and checked the resulting identities independently. Dividing initial forms by their common monomial, using a power of the product of the torus coordinates as the right-hand side, and halving exponents of variables that occur only evenly substantially reduced several computations. Some cases instead used split trees or structural elimination replays. Section 5 describes the formats and their mathematical justification.

The companion ledger.

Fourteen Laurent-identity batches in `companion/results/` record generator indices, cell weights, monomial shifts, multipliers, and a nonzero mass factor q . Each identity has the form

$$\sum_j c_j r^{-e_j} \text{in}_{w_C} f_{i_j} = q(a, b, c) r^d.$$

The checker expands both sides exactly, checks that the recorded weight is the sum of the cell rays, and verifies support constancy on the cell. The right-hand side is a Laurent unit at generic masses. The union also includes a structural ratio-elimination proof for cell 346 and its symmetry images, whose positive-mass branches are checked by exact Sturm arithmetic; conditional transfers from cell 325, excluded away from the mass equalities $a = 4b$ or $b = 4a$; and coefficient-level symmetry transports. The replay reconstructs the union of these proofs and verifies that it is exactly the 10,965-cell ledger. Overlapping batches contribute only once. The enumerated cell assignments are in `data/cone_accounting.json`; this ledger is checked evidence, not a list of search successes.

The final three exclusions.

Direct lifting remained expensive for representatives 81, 121, and 206. For each representative the final argument uses Lemma 5 at $(a, b) = (3, 7)$, $p = 32003$. Each star has 60 cones, including its target. All 37 initial forms of $F \cup \{G, H\}$ are constant, U -free, and retain every coefficient at this specialization. The neighboring-cone checks use modular cofactor identities, except for cluster cones 2174, 2885, and 3494, which use a short exact elimination replay. The target-cell checks use the linear even-variable reduction and the bivariate leaf argument described below. The mass-swapping symmetry also excludes their partners 316, 50, and 220.

Coverage and residual.

Table 1 gives disjoint counts, assigning each cone to the first applicable certificate in the checked ledger. Symmetry images under the pair swaps are included in their source rows; the additional transport row records the remaining four mass-exchange images. The 96 single-term virial cells overlap earlier rows, leaving 57 in that row and the separate scaling ray. Within each row, pair

TABLE 1. Disjoint accounting of the nonzero support-fan cones.

Certificate or residual class	Cones
Companion Laurent and structural identities	10,965
Other Laurent identities	5,258
Resultant specialization	8
Split trees	6
Structural elimination	25
Modular star criterion	16
Additional mass-exchange transport	4
Single-term virial initial forms	57
Scaling ray	1
Excluded total	16,340
Residual	98
Total	16,438

TABLE 2. Certificate representatives and expansion for Table 1.

Class	Representatives and check in Section 5	Cones
Companion	Fourteen batches, structural and conditional transfers; companion ledger above	10,965
Laurent	1,562 representatives in <code>cert</code> , <code>dcert</code> , <code>gcert</code> , <code>hcert</code> , <code>ecert</code> ; unit identities	5,258
Resultant	227, 2440; two-variable resultant specialization	8
Split trees	378, 427; binary split identities	6
Elimination	334, 1767; 49, 2171, 2174, 2175; mass-exchange partners 2345, 218, 2885, 3494, 3545	25
Modular stars	81, 121, 206; partners 316, 50, 220	16
Transport	Remaining mass-exchange images from 353, 3097	4
Single-term G	23 representatives; enumerated accounting and Lemma 9	57
Scaling ray	79; Lemma 9	1

swaps give orbits of sizes one, two, or four. The enumerated accounting records every representative,

target cell, orbit size, and extra mass exchange; preceding rows take priority. In particular the two split representatives contribute $2 + 4$ cones, the two resultants $4 + 4$, the eleven elimination representatives $1 + 2 \cdot 4 + 8 \cdot 2$ with the actual multiplicities recorded in the ledger, and the six modular representatives $2 + 4 + 2 + 2 + 4 + 2$.

The residual has 77 distinct primitive rays. Every one has positive pairing with

$$(3) \quad \eta = (-23, 1, 1, 51, 1, 1, 51, 24, 13, 13).$$

The checker reconstructs the residual from the certificate coverage and verifies these pairings in integer arithmetic.

Interpretation and controls.

A weight w records possible distance growth orders $r_{ij} \sim t^{-w_{ij}}$. The residual cones are candidate asymptotic distance patterns consistent with the supports and not removed by our certificates. They are not classified as actual degenerations of real configurations. Their common half-space is sufficient: no positive-dimensional tropical variety can be supported there. The elementary example following Lemma 5 supplies an exact negative control for the modular argument: its target specialization is empty, but the adjacent cell has torus zeros. The checker in `verifier/review_checks.py` confirms both statements and also confirms why a test only over $\mathbb{F}_p$ is insufficient.

5. CERTIFICATES AND INDEPENDENT VERIFICATION

Format.

`data/exact/cert/k.cof` contains:

- the cone index k ;
- the subset $\mathcal{P}$ (0-based indices into `data/sys.json`);
- the ring variables (kept coordinates and the auxiliary variable, named `u` in the files);
- for each generator in order, the terms of its cofactor C_g as
`numerator|denominator|exponent vector`,
 where numerator and denominator are in $\mathbb{Z}[a, b]$.

Verifier.

`verifier/verify.py` uses python-flint's FLINT exact multivariate arithmetic [20] and does not use SINGULAR or sympy. It

- (1) parses `fan.out` and `sys.json` itself;
- (2) checks the hypothesis of Lemma 3 for every $g \in \mathcal{P}$;
- (3) recomputes $\mathcal{V}$ and checks surjectivity onto the fixed coordinates $\mathcal{D}$;
- (4) clears the denominators into $\ell = \text{lcm}$ and verifies the following identity in exact arithmetic:

$$\sum (\ell C_g) \bar{g} + (\ell C_0) (1 - \zeta \prod r) = \ell.$$

The following variants are also accepted.

- **system F+G**: the certificate refers to the augmented system $F \cup \{G\}$ in the 11 coordinates (r_{ij}, U) , where $G = \sum_{i < j} m_i m_j r_{ij}^2 - U$ has index 35. U has weight 0 because its fixed value is a nonzero constant with trivial valuation (Remark 10). The verifier builds G itself from the masses $(a, a, b, b, 1)$. This is used for the scaling cone 79, for the 96 cones on which $\text{in}_w G$ is a single term (a one-term certificate $C_G = 1/\text{coeff}$), and for representatives whose unit test over $\mathbb{Q}(a, b)$ timed out without G .
- **divmono**: each $\bar{g}$ is divided by the greatest common monomial of its terms before the identity is checked. Monomials are units of the Laurent ring, so this does not change the torus zero set.

- **even** $v_1, \dots$: each listed variable occurs in every $\bar{g}$ only with even exponents, and the verifier halves those exponents, i.e. writes $\bar{g}(r) = \bar{g}'(r_v^2)$. The map $r_v \mapsto s_v = r_v^2$ is a surjective endomorphism of the torus. A Laurent identity $\sum C'_i \bar{g}'_i = s^N$ therefore pulls back to $\sum C'_i(r^2) \bar{g}_i = r^{2N}$, which is again a unit certificate for $\langle \bar{g} \rangle$.
- **rhs prod^N**: no Rabinowitsch variable. The identity checked is $\sum_g (\ell C_g) \bar{g} = \ell \cdot (\prod_{j \in \mathcal{I}} r_j)^N$ in $\mathbb{Q}[a, b][r_{\mathcal{I}}]$, where $\mathcal{I}$ is the set of kept coordinates. On the torus the right-hand side vanishes nowhere (for $\ell(a, b) \neq 0$), so $V(\text{in}_w F) \cap T = \emptyset$, exactly as in the Rabinowitsch form.

Denominators in the certificates must lie in $\mathbb{Z}[a, b]$ (the verifier checks this). The identity therefore holds in $\mathbb{Q}[a, b][r, \zeta]$ after multiplying by ℓ . It specializes to every mass pair with $\ell(a, b) \neq 0$, which is a nonempty Zariski-open (generic) condition, and the finitely many polynomials ℓ are recorded in the verifier output.

Split-tree certificates.

For a few cones the single lift identity is too large to compute, although the ideal splits naturally. Typically one initial form factors as $g_1 g_2$ modulo a binomial of the system. For cone 378, for example,

$$a^2 + ab r_{13}^2 + ab r_{24}^2 + b^2 r_{34}^2 \equiv (a + b r_{13}^2)(a + b r_{24}^2)$$

modulo $r_{34}^2 - r_{13}^2 r_{24}^2$. A split-tree certificate (`exact/split/<k>.split`, checker `verifier/verify_split.py`) is a finite binary tree. A node v with path ideal $\mathfrak{J}_v = \mathfrak{J} + (p_1, \dots, p_s)$ (the base Rabinowitsch ideal $\mathfrak{J}$ of `verify.py` plus the polynomials chosen at the ancestors) is either a *leaf*, carrying cofactors with $\sum_i C_i k_i = \ell_v \in \mathbb{Q}[a, b] \setminus \{0\}$, or a *split* node with polynomials q_1, q_2 and cofactors with $\sum_i C_i k_i = \ell_v q_1 q_2$. Both children $\mathfrak{J}_v + (q_1)$ and $\mathfrak{J}_v + (q_2)$ must be certified. Over $\mathbb{Q}(a, b)$, if $q_1 q_2 \in \mathfrak{J}_v$ and $1 = x_1 + c_1 q_1 = x_2 + c_2 q_2$ with $x_j \in \mathfrak{J}_v$, then

$$1 = (x_1 + c_1 q_1)(x_2 + c_2 q_2) \in \mathfrak{J}_v.$$

By induction $1 \in \mathfrak{J}$, which is exactly the statement certified by a `.cof` file. The support and torus-slice checks are shared with `verify.py` (they are run on the header). All identities are checked exactly with `python-flint`, and the generic condition is that no ℓ_v vanishes.

Elimination certificates (cones 334, 1767 and 49, 2171, 2174, 2175).

These cones are certified by an explicit elimination that the checkers `verifier/verify_334.py` and `verifier/verify_49.py` replay in exact arithmetic (`python-flint`, no `SINGULAR`). In both groups six initial forms $g_1, \dots, g_6$ in five variables suffice, and they are identical for all cones of the group. The checkers recompute them from the fan through the support and torus-slice checks of `verify.py`. The argument is pointwise. Let z be a common zero of the g_i on the torus over the algebraic closure K of the coefficient field, with the masses algebraically independent. Each derived polynomial below vanishes at z , and every branch ends in a contradiction.

- (1) Three of the forms are affine-linear in three squares; by Cramer's rule $D r^2 - N_r$ is an explicit combination of them, and D is a monomial. Replacing squares and squaring away the linear unknowns gives division-free polynomials $E_1, \dots, E_{14}$ in two variables (x, y) .
- (2) Polynomials are divided only by factors known to be nonzero at z : monomials, the masses, and the factors X_q, Y_q, S_q of N_r (nonzero because $D r^2 = N_r$ and $r \neq 0$). The remaining factors F_i (the Cramer determinant of the second linear system) are treated as separate branches $F_i(z) = 0$.
- (3) Resultants lie in the ideal of their arguments, so they vanish at z . The final non-vanishing $\text{Res}_y(R', R'') \neq 0$ is certified by specialization: at one rational parameter point modulo a prime the relevant degrees are preserved and the specialized gcd is 1. As in `verify_res.py`, this proves that the resultant is a nonzero element of $\mathbb{Z}[a, b]$. The common factors that are left are explicit. For 49 it is the irreducible cubic $\phi = (ay + 1)^3 - b^2(ay^3 + 1)$. Over the field

$\mathbb{L} = \mathbb{Q}(a, b)[y]/(\phi)$ the checker exhibits $\psi = -\beta_0/\beta_1$ (where $\beta_1 x + \beta_0$ is the first subresultant of X_q, Y_q). It verifies $X_q(\psi) = 0$ and $(x - \psi)^2 \mid \widehat{E}_{10}, \widehat{E}_{13}$ in $\mathbb{L}[x]$, and $\deg \gcd_{\mathbb{L}}(\widehat{E}_{10}, \widehat{E}_{13}) \leq 2$. The degree bound holds because the principal subresultant coefficient is nonzero at a specialization, and Gauss's lemma applies since ϕ is primitive. So $x(z) = \psi$ and $X_q(z) = 0$, a contradiction.

For 334 all six forms are homogeneous of degree one in (a, b) ; divide them by $b \neq 0$ and replace a by a/b . This is an equivalence on the mass torus and justifies $b = 1$, rather than a specialization to a possibly exceptional fiber. The diagonal $r_{23} = r_{24}$ is closed by a resultant chain in $\mathbb{Z}[a]$. Off the diagonal, the symmetry $(r_{13} r_{14})(r_{23} r_{24})$ of $g_1, \dots, g_5$ (not of g_6 , which enters only through norms $E \cdot \sigma E$) reduces to $s = r_{23} + r_{24}$, $p = r_{23} r_{24}$. The only common zero left, $p = 3/(a^2 - 1)$, $s = -3a/(a^2 - 1)$, has $X_q Y_q S_q = 0$. The six exact input polynomials and the division-free definitions of the derived polynomials are given in Appendix A. The irreducibility and specialization steps are justified there as well.

Dehomogenized computations.

For some cones every generator of the chosen subset is homogeneous in the masses (a, b) (the mass torus acts on the subsystem). The cofactors are then computed over $\mathbb{Q}(a)$ at $b = 1$ and rehomogenized: if

$$\sum_i C_i(a) \bar{g}_i(a, 1) = 1$$

and $\bar{g}_i$ is homogeneous of degree d_i in (a, b) , then

$$\sum_i b^{-d_i} C_i(a/b) \bar{g}_i(a, b) = 1$$

(`exact/ecert/rehom.py`). The resulting file is an ordinary certificate over $\mathbb{Q}(a, b)$ and is checked by `verify.py` without any reference to this construction.

Independent support-fan verification.

The `gfan` [18] computations with versions 0.7 and 0.6.2 agree on primitive rays and cones. This is a cross-version replication, not an independent implementation. The version-0.7 build record lists portability changes to memory-resource headers, integer overloads, and execution policies; the original patch diff is not retained. Our separate check does not rely on those changes.

First, `verifier/fan_structure.py` independently reconstructs every one of the 16,438 nonzero cones as a cone of the common refinement of the Newton normal fans. For each support it intersects the maximizing faces on the cone rays, obtaining a face T_i with at least two terms. Its normal region is given by the equalities within T_i and the inequalities $\langle w, m_0 - m \rangle \geq 0$, $m_0 \in T_i$, for the other terms. Exact rational nullspace arithmetic restricts these conditions to the cone's span, of dimension at most four. Enumerating independent sets of $d - 1$ generating rays gives every facet normal. Each facet inequality occurs among the defining inequalities of the normal region, proving equality of the stored cone with that region. The checker also verifies that the rays are primitive and extreme, lineality is zero, and every facet is a stored face: 44,998 facet incidences in total. Normal-fan cones intersect in common faces; closure under facets gives closure under all faces. Thus relative interiors partition the support, and the nearby-cone face relation used in Lemma 5 holds independently of `gfan`. The checker also verifies invariance of both all 35 supports and the complete stored fan under the three generating relabelings.

For completeness of the support, `cddlib` with GMP [21] converts the 3,120 inclusion-maximal cones to exact rational half-spaces. A rational-linear solver asks whether a weight outside their union lies in all thirty-five tropical hypersurfaces. For each support $\mathcal{A}_i$ it introduces a maximum t_i and imposes

$$t_i \geq \langle w, m \rangle \quad (m \in \mathcal{A}_i), \quad \#\{m \in \mathcal{A}_i : t_i = \langle w, m \rangle\} \geq 2.$$

Membership in the stored cones is negated using their exact half-spaces. Homogeneity allows $\max_j |w_j| = 1$. The verified order-eight symmetry action has three coordinate orbits represented by r_{12}, r_{13}, r_{15} , so it suffices to check their positive and negative normalized cases. For r_{12} fixed, its stabilizer can place the largest of the four cross-pair coordinates at r_{13} ; for r_{13} fixed it can arrange $w_{12} \geq w_{34}$; for r_{15} fixed it can arrange $w_{13} \geq w_{14}$. These inequalities reduce search without removing any symmetry orbit.

All six cases returned UNSAT independently in Z3 5.1.0 [22] and Yices 2.7.0 [23], using exact rational arithmetic. The Yices replay uses standard SMT-LIB QF_LRA: at least one term attains the maximum, and every attaining term requires a different attaining term. This Boolean encoding is independent of Z3's pseudo-Boolean operator. The half-spaces were regenerated and matched the checked inputs exactly. Since a nonempty difference of these rational polyhedral sets has a rational point, this proves containment for all real weights. Thus the stored fan has exactly the prevariety support required by Lemma 5. The audit code, input hashes, checked base query, and per-case results are distributed with the verifier. As a control, it proves completeness for the three-ray tropical line of $1 + x + y$ and finds a witness when one ray is removed. A timeout or UNKNOWN is never accepted as a completeness result.

Modular specialization certificates.

For each of the three targets $k = 81, 121, 206$, `verify_lemS.py` computes a star superset directly from `fan.out`, selecting cones whose ray sets contain the target rays. Every true coface is included; checking any extra cones is harmless. The independently verified fan structure justifies using this superset. It requires all 37 initial forms to be constant and U -free, verifies that every coefficient remains nonzero modulo 32003 at $(3, 7)$, and checks every cone of the star. This explicitly enforces the intersection $\Phi_C \cap \Phi_D$ required by Lemma 5. Most neighboring cones have ordinary modular cofactor identities in `lemS/cert/`. The cluster cones 2174, 2885, and 3494 instead use `cluster_mod.py`: it reconstructs six forms, derives three square relations by Cramer's rule, replays division-free elimination to fourteen bivariate polynomials, and checks that their common roots force a zero torus coordinate. The transported cluster forms are reconstructed at their actual cones.

Target-cell replays.

The checker `verify_lemS_rep.py` reconstructs all target forms, validates its homogeneous torus slice, and replaces the three even variables by their squares. One form eliminates a squared coordinate; a second gives

$$c_{23}(x, y)s_{23} + c_{24}(x, y)s_{24} = 0.$$

Outside $c_{23} = c_{24} = 0$, put $s_{23} = \kappa P$, $s_{24} = \kappa Q$, with κ, P, Q and the eliminated squared coordinate nonzero. Homogeneous equations then supply forms independent of κ . Three of them determine S^2, T^2, W^2 ; two cube equations make $1/T$ and $1/W$ affine in $1/S$. Squaring the first relation determines $1/S$ rationally. Every denominator is checked as nonzero on the branch; for cell 206 an additional bivariate leaf proves the required denominator cannot vanish. Substitution into further initial forms gives four bivariate polynomials. Their common zero set is checked exactly outside the explicitly excluded factors. Cases where a cube-relation coefficient vanishes are handled separately by four further leaves, so division does not discard a branch.

The exceptional branch $c_{23} = c_{24} = 0$ gives $(x^3, y^3) = (1, 1)$ for 81 and 121, and $(1, -1/13)$ for 206. Since $32003 \equiv 2 \pmod{3}$, a cube root of the nonzero constant lies in $\mathbb{F}_{32003}$, and all roots lie in its quadratic extension. Each branch has nine root pairs. The generator `b0_generate.py` found small cofactor identities for these cases. The independent checker `b0_verify.py` reconstructs their input polynomials and verifies all nine identities in $\mathbb{F}_{32003}[t]/(t^2 + t + 1)$. The branch check is mandatory; a partial replay cannot certify a cell.

Bivariate leaf check.

The helper `leafchk.py` computes a nonzero resultant in x of two of the input polynomials. Every common affine root has a y -coordinate among the roots of this resultant. For each irreducible factor, the checker forms the corresponding finite extension field, specializes all input polynomials, and computes their univariate gcd in x . It removes only factors belonging to the explicitly excluded polynomials. A constant gcd in every fiber proves the absence of a permitted common root. Extra variables are rejected. This is finite-field exact arithmetic and does not call SINGULAR.

Coverage and pointedness.

The complete replay reconstructs the cones excluded by actual identities and elimination branches, then checks (3) on every residual ray. The companion replay checks its Laurent, structural, and symmetry proofs against the 10,965-cell union. Neither search output nor cached success markers establishes an exclusion.

Trusted arithmetic and solvers.

The independent replay relies on Python, the verifier code, and FLINT through `python-flint` for exact rational, polynomial, and finite-field operations; the companion identities additionally use SymPy's exact expansion. The support-completeness audit relies on `cddlib/GMP`, `Z3`, and Yices rational SMT solving. The recorded `Z3` version is 5.1.0. Singular generates characteristic-zero identities but is not used to verify them. Gfan generates the initial fan; the normal-region and facet checks independently verify its structure, while the solver audit verifies completeness. These are executable exact checks with stated software dependencies, not a proof-assistant formalization.

6. CODE AND CERTIFICATE DATA

The source, verifier, and certificate archive are versioned at

<https://github.com/brandonlign/tropcc/tree/v1.2.1>

The repository is private; access must be granted for independent replay. A public archival deposit with a persistent identifier remains necessary before submission. The versioned certificate archive is attached to release `v1.2.1`; its SHA-256 digest and all extracted file hashes are recorded in `certificate_manifest.json`. No archival DOI has been assigned. The repository README gives installation and full replay commands; `CERTIFICATES.md` maps claims to checks. The stable filenames `verify_lemS.py` and `verify_lemS_rep.py` refer to the modular-star criterion, now Lemma 5.

APPENDIX A. EXPLICIT ELIMINATION INPUTS AND DERIVED POLYNOMIALS

We give finite algebraic definitions rather than expanding the large resultant coefficients. Together with the branch prescriptions below, these determine every polynomial used in the two elimination replays. The six input forms are also stored in `data/elimination_forms.json`, and the independent reconstruction checks their agreement with F and the stated cones.

Cone group 334.

Put $(A, B, T, x, y) = (r_{13}, r_{14}, r_{34}, r_{23}, r_{24})$ and take $b = 1$ after the homogeneous mass reduction above. The six forms, in their checked order, are

$$\begin{aligned} g_1 &= -A^3 B^3 a - A^3 \\ &\quad + AB^3 a(x-1)(x+1) - AB^2 \\ &\quad + AT^2 - 2B^3 \end{aligned}$$

$$\begin{aligned}
g_2 &= -A^3B^3a + A^3Ba(y-1)(y+1) \\
&\quad - 2A^3 - A^2B \\
&\quad - B^3 + BT^2 \\
g_3 &= A^2y^3 + B^2x^3 \\
&\quad - 2ax^3y^3 - x^3y^2 - x^3 - x^2y^3 - y^3 \\
g_4 &= A^2axy^3 + T^2x \\
&\quad - ax^3y^3 - axy^3 - x^3 - xy^2 - 2y^3 \\
g_5 &= B^2ax^3y + T^2y \\
&\quad - ax^3y^3 - ax^3y - 2x^3 - x^2y - y^3 \\
g_6 &= -2B^3T^3a - B^3T^2y \\
&\quad + B^3y(x-y)(x+y) - B^2T^3ay \\
&\quad - T^3ay(y-1)(y+1)
\end{aligned}$$

The same forms occur at cone 1767. The variables A, B, T in this appendix are local distance variables; they are unrelated to A_{ij} .

Cone group 49.

Put $(A, B, T, x, y) = (r_{14}, r_{15}, r_{45}, r_{34}, r_{35})$. Here both parameters remain. The six forms are

$$\begin{aligned}
g_1 &= -A^3B^3b - A^3 \\
&\quad + AB^3b(x-1)(x+1) - AB^2 \\
&\quad + AT^2 - 2B^3b \\
g_2 &= -A^3B^3b + A^3Bb(y-1)(y+1) \\
&\quad - 2A^3 - A^2Bb \\
&\quad - B^3b + BT^2b \\
g_3 &= A^2by^3 + B^2x^3 \\
&\quad - 2ax^3y^3 - bx^2y^3 - by^3 - x^3y^2 - x^3 \\
g_4 &= A^2axy^3 + T^2x \\
&\quad - ax^3y^3 - axy^3 - 2by^3 - x^3 - xy^2 \\
g_5 &= B^2ax^3y + T^2by \\
&\quad - ax^3y^3 - ax^3y - bx^2y - by^3 - 2x^3 \\
g_6 &= -2B^3T^3b - B^3T^2by \\
&\quad + B^3by(x-y)(x+y) - B^2T^3ay \\
&\quad - T^3ay(y-1)(y+1)
\end{aligned}$$

They agree at cones 49, 2171, 2174, and 2175. The two groups use different initial subsets and distance slices, explaining the different mass coefficients in g_1, g_2, g_6 . Their forms g_3, g_4, g_5 agree at $b = 1$, so the X_q, Y_q, S_q used for group 334 are exactly the $b = 1$ specializations of those defined below for group 49.

A common division-free construction.

In either group write

$$(g_3, g_4, g_5)^\top = M(A^2, B^2, T^2)^\top - v,$$

where M, v are obtained by coefficient extraction from the displayed forms. Put $d = \det M$ and let n_A, n_B, n_T be its Cramer numerators. Thus

$$d(A^2, B^2, T^2)^\top - (n_A, n_B, n_T)^\top = \text{adj}(M)(g_3, g_4, g_5)^\top.$$

The determinant d is a nonzero monomial, so is a unit on the generic mass and distance torus. For a polynomial P , define $\rho(P)$ termwise: replace each $A^{2i+e}B^{2j+f}T^{2k+h}$ by $n_A^i n_B^j n_T^k A^e B^f T^h d^{N-i-j-k}$, where $e, f, h \in \{0, 1\}$ and N is the largest $i+j+k$ among its terms. On a common zero, $\rho(P) = d^N P$. Let prim remove integer content and the greatest common monomial, including mass monomials. These factors are nonzero on the generic torus. Set $R_i = \text{prim } \rho(g_i)$ for $i = 1, 2, 6$ and extract coefficients by

$$R_1 = A(a_1 B + b_1) + c_1 B,$$

$$R_2 = B(a_2 A + c_2) + b_2 A,$$

$$R_6 = T(a_6 B + b_6) + c_6 B.$$

All lowercase coefficients here are polynomials in x, y over the mass ring, not mass parameters. Applying $\text{prim } \rho$ to $c_1^2 B^2 - A^2(a_1 B + b_1)^2$, $b_2^2 A^2 - B^2(a_2 A + c_2)^2$, and $c_6^2 B^2 - T^2(a_6 B + b_6)^2$ gives, respectively,

$$\alpha B + \beta, \quad \gamma A + \delta, \quad \alpha_6 B + \beta_6.$$

Next, take $\text{Res}_A(R_1, R_2)$ and $\text{Res}_B(R_1, R_2)$, remove their maximal factors B and A , respectively, then apply $\text{prim } \rho$. The results are $\mu B + \omega_0$ and $\mu_2 A + \omega_2$. Define the following polynomials, applying prim to each right-hand side:

$$\begin{aligned} E_1 &= \beta^2 d - \alpha^2 n_B, & E_2 &= \delta^2 d - \gamma^2 n_A, \\ E_3 &= \alpha_6 \beta - \alpha \beta_6, \\ E_4 &= a_1 \beta \delta - b_1 \alpha \delta - c_1 \gamma \beta, \\ E_5 &= a_2 \delta \beta - c_2 \beta \gamma - b_2 \delta \alpha, \\ E_6 &= (b_1 \delta)^2 d - (c_1 \gamma - a_1 \delta)^2 n_B, \\ E_7 &= (b_2 \delta)^2 d - (c_2 \gamma - a_2 \delta)^2 n_B, \\ E_8 &= (c_1 \beta)^2 d - (b_1 \alpha - a_1 \beta)^2 n_A, \\ E_9 &= (c_2 \beta)^2 d - (b_2 \alpha - a_2 \beta)^2 n_A, \\ E_{10} &= \omega_0^2 d - \mu^2 n_B, & E_{11} &= \mu \beta - \alpha \omega_0, \\ E_{12} &= \mu \beta_6 - \alpha_6 \omega_0, & E_{13} &= \omega_2^2 d - \mu_2^2 n_A, \\ E_{14} &= \mu_2 \delta - \gamma \omega_2. \end{aligned}$$

Each E_i vanishes at a common torus zero; these constructions use multiplication, squaring, resultants, and division only by known units.

Factors and branches for group 49.

The factors which must be nonzero are

$$\begin{aligned} X_q &= a^2 x^4 y^4 + a b x^3 y^4 + a b x y^4 + b^2 y^4 - x^4, \\ Y_q &= a^2 x^4 y^4 + a x^4 y^3 + a x^4 y - b^2 y^4 + x^4, \\ S_q &= a^2 x^4 y^4 - b^2 y^4 - b x^3 y - b x y^3 - x^4. \end{aligned}$$

Indeed $n_A = -2x^3 X_q$, $n_B = -2by^3 Y_q$, $n_T = 2ax^3 y^3 S_q$. Remove their maximal powers from E_i to obtain $\tilde{E}_i$. On the leaf also remove powers, while the quotient is nonconstant, of

$$ax^2 y^2 \pm by^2 \pm x^2 \quad (\text{all four sign choices}), \quad ax^3 y^3 + by^3 + x^3,$$

to obtain $\widehat{E}_i$; the zero loci of these five factors are checked separately using $\widetilde{E}_1, \widetilde{E}_2, \widetilde{E}_3$. Thus no division discards a branch. The leaf uses $\text{Res}_x(\widehat{E}_{10}, \widehat{E}_{13})$ and $\text{Res}_x(\widehat{E}_{10}, \widehat{E}_{12})$ as described in Section 5.

Irreducibility and the specialization bound.

Let $\phi = (ay + 1)^3 - b^2(ay^3 + 1)$. Over $\mathbb{Q}(a, y)$ the quotient $(ay + 1)^3/(ay^3 + 1)$ is not a square: its order at $y = -1/a$ is three, since the denominator there is $1 - a^{-2} \neq 0$. Hence ϕ , viewed as a quadratic in b , is irreducible over $\mathbb{Q}(a, y)$. Its two coefficients are coprime in $\mathbb{Q}(a)[y]$, so Gauss's lemma gives irreducibility in $\mathbb{Q}(a)[y, b]$. As a polynomial in y it is primitive over $\mathbb{Q}(a)[b]$ (its y -coefficient is the unit $3a$ there), giving irreducibility over $\mathbb{Q}(a, b)$. Therefore $\mathbb{L} = \mathbb{Q}(a, b)[y]/(\phi)$ is a field.

A principal subresultant coefficient is polynomial in the coefficients of its two arguments. If it vanished in $\mathbb{L}$, its polynomial numerator would be divisible by ϕ after clearing coefficient denominators. Primitivity in $\mathbb{Z}[a, b][y]$ follows because the coefficients $3a$ and $1 - b^2$ are coprime over $\mathbb{Q}[a, b]$; Gauss's lemma makes the divisibility valid before specializing. It would then vanish at every modular root of the specialized ϕ for which denominators survive. The checked specialization preserves the x -degrees and has gcd degree two, so its degree-two principal subresultant coefficient is nonzero. This proves the generic bound $\deg \gcd_{\mathbb{L}}(\widehat{E}_{10}, \widehat{E}_{13}) \leq 2$. Together with the verified double root $x = \psi$, it identifies the gcd and yields $X_q = 0$, a contradiction.

Branches for group 334.

The diagonal $x = y$ first gives $B = \pm A$. The negative sign forces $T = 0$. For the positive sign put $(A, B, x, y, T) = (z, z, t, t, v)$ and remove the common monomial from each specialized g_i , obtaining P_i . Choose signs of P_1, P_4 so their v^2 coefficients are $-1, +1$. Define

$$\begin{aligned} Q_1 &= \text{prim Res}_v(P_6, P_4), \\ Q_2 &= \text{prim Res}_z(P_1 + P_4, P_3), \\ Q_3 &= \text{prim Res}_z(Q_1, P_3). \end{aligned}$$

The exact computation gives $0 \neq \text{Res}_t(Q_2, Q_3) \in \mathbb{Z}[a]$, excluding a common zero at generic a . Off the diagonal let $s = x + y$, $\pi = xy$ and use the swap $(A \ B)(x \ y)$. For $1 \leq i \leq 14$, $i \neq 3, 12$, take the symmetric part $E_i + \sigma E_i$ and the divided antisymmetric part $(E_i - \sigma E_i)/(x - y)$; for $i = 3, 12$ take the norm $E_i \sigma E_i$. Rewrite each exactly in (s, π) , verifying back-substitution. The branch factors are

$$\begin{aligned} a\pi^2 - s^2 + 2\pi, \quad a\pi^2 + s^2 - 2\pi, \\ a\pi^3 + s^3 - 3s\pi, \quad a^2\pi^4 - s^4 + 4s^2\pi. \end{aligned}$$

The checker treats their zero loci separately. On the leaf it uses the symmetric and antisymmetric parts of E_{10} and the norm of E_{12} . Write $\widehat{E}_{10s}, \widehat{E}_{10a}, \widehat{E}_{12n}$ for the leaf quotients of the symmetric, divided antisymmetric, and norm polynomials, after removing only the nonzero branch factors. Put

$$R_1 = \text{Res}_s(\widehat{E}_{10s}, \widehat{E}_{10a}), \quad R_2 = \text{Res}_s(\widehat{E}_{10s}, \widehat{E}_{12n}), \quad P_* = (a^2 - 1)\pi - 3.$$

Remove the a -content and all factors π, P_* from R_1 , leaving R'_1 . Preserved degrees and the coprime specialization $a = 7$ modulo 1000003 give $\text{Res}_\pi(R'_1, R_2) \neq 0$ over $\mathbb{Q}(a)$. Hence a common zero has $\pi = 0$ or $P_* = 0$. The torus excludes $\pi = 0$. The exact factorization further gives

$$\begin{aligned} \text{prim}_s \gcd(\text{Res}_\pi(\widehat{E}_{10s}, P_*), \text{Res}_\pi(\widehat{E}_{10a}, P_*)) &= L_*^e, \\ L_* &= (a^2 - 1)s + 3a, \end{aligned}$$

up to a unit in $\mathbb{Q}(a)$, with $e > 0$; prim_s removes parameter content. For $T_* = n_A n_B n_T$ in (s, π) , the checker verifies $L_* \mid \text{Res}_\pi(T_*, P_*)$. Linearity of P_* forces $T_* = 0$ at $P_* = L_* = 0$, impossible on the torus. The mass exception $a^2 = 1$ is included when spreading.

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